

"Design engineers are the most sought after because of the industry's emphasis on continuous new product development, miniaturisation and innovation in integration. Typically, a graduate with a master's degree can expect about 10 per cent higher than the one with a bachelor's degree, and someone with a PhD degree can expect a wide range," says C.P. Ravikumar, technical director-university relation, Texas Instruments.

Based on continual performance, knowledge and competence, you can expect steep growth in remuneration as well as roles and responsibilities. The growth prospects for professionals in Indian companies are definitely promising given the scarcity of talent. For example, a young engineer can easily grow to become a team leader or project manager in a span of six to nine years. The ability to handle multitasking jobs may be a defining factor. As you gain experience, this field offers one of the best growth potentials, both on the technical as well as management front. Though the global recession has hugely affected starting salaries in the private sector at present, the previous year's industry feedback reveals an average pay package between Rs 700,000 and Rs 1.5 million per annum for a four-to five-year-experienced professional.

How to get chosen?

"For VLSI design, companies prefer BE/B.Tech, ME/M.Tech and PhD holders with specialisation in electronics, telecommunications and electrical VLSI domains. Typically, these professionals should have hands-on

An aspirant in this field is expected to know the physics of semiconductor devices, linear systems, probability and random variables, engineering mathematics, circuit analysis and engineering electromagnetics.

—Dr S.Karthik, Engineering Director, India Product Development Centre, Analog Devices India



experience in systems design, digital application-specific integrated circuit (ASIC) design, physical design, mixed-signal IC design, VHDL or VHSIC (very high-speed integrated circuit), VLSI design, circuit design and simulations, microcontrollers, digital PCB design and routing. You can expect an engineer with M.Tech/PhD degree and VLSI skills to have an edge in the job market," explains Bhagat.

For working in this field, it is mandatory to have in-depth knowledge of either electronics, electrical or computer science. Though an engineering degree is the obvious qualification to earn, graduate and postgraduates in physics also qualify to become an engineer. The physics of semiconductor devices is the fundamental basis of VLSI.

"The subject of VLSI design is quite vast. It is not fair to expect both breadth and depth from a fresh recruit. At the undergraduate level, we expect a good understanding of fundamental concepts. Our campus recruitment is based on a written test and an inter-

view. The written test is separate for hardware and software engineers. The hardware paper has a section on aptitude, a section on digital design and a section on analogue design. We ask the students to select any one section between digital and analogue design. The software paper has an aptitude section and a software section. Candidates who fare well in the written test are shortlisted for interview. From my experience, the interviews are mostly based on problem-solving exercises and test the conceptual understanding rather than rote learning. We don't pay too much importance to the B.Tech project done by the student," says Ravikumar.

"For postgraduate students, our process is similar, except that we may go into their thesis. For candidates with a PhD, we also ask for a presentation where we invite our experts," he adds.

The earlier, the better. If you ever tinkered with a broken radio set, you have already started. Academically, the right time to acquaint yourself with this niche field of electronics is when you are in the second or third year of engineering.

Dr Karthik points out, "An aspirant in this field is expected to know the physics of semiconductor devices, linear systems, probability and random variables, engineering mathematics (Fourier, Laplace and Z transforms), circuit analysis and engineering electromagnetics. For hiring a fresher, we look for in-depth understanding of

Component-level knowledge of electronic circuits and devices and thorough understanding of network theory, linear and digital design concepts are must-have.



—Vivek Pawar, CEO, Sankalp Semiconductor